
9 (a)(i)

Magnetic flux through the surface is defined as the product of the magnetic flux density B **perpendicular** to the surface and the area A of the surface. [B1]

Or

Magnetic flux is the product of the magnetic flux density and the area **normal** to the lines of flux. [B1]

9 (a)(ii)

Faraday's law of electromagnetic induction states that the magnitude of the induced e.m.f. across a conductor is directly proportional to the rate of change of magnetic flux linkage (or the rate of cutting of magnetic flux). [B1]

Lenz's Law states that the induced e.m.f in a closed circuit will produce effects that oppose the magnetic flux change that causes it.

OR

Lenz's Law states that the induced current in a closed circuit will flow in such a way as to oppose the change that causes it.

[B1]

9 (b)(i).

Magnetic flux linkage $\Phi = NBA$ [C1]

At 30.0 mm position,

$$\begin{aligned}\Phi_1 &= (50)(1.0 \times 10^{-3})(16.0 \times 10^{-4}) \\ &= 8.0 \times 10^{-5} \text{ Wb}\end{aligned} \quad [\text{A1}]$$

At 10.0 mm position,

$$\begin{aligned}\Phi_2 &= (50)(32.0 \times 10^{-3})(16.0 \times 10^{-4}) \\ &= 2.56 \times 10^{-3} \text{ Wb}\end{aligned} \quad [\text{A1}]$$

9 (b)(ii)

From Faraday's law,

$$E = \left| \frac{\Delta \Phi}{\Delta t} \right| \quad [\text{M1}]$$

$$0.80 \times 10^{-3} = \frac{2.56 \times 10^{-3} - 8.0 \times 10^{-5}}{\Delta t}$$

time take $\Delta t = 3.10 \text{ s}$

average speed = $(30.0 - 10.0) / 3.10$

$$= 6.45 \text{ mm s}^{-1}$$

$$= 6.5 \text{ mm s}^{-1} \text{ (2s.f.)} \quad [\text{A1}]$$

9 (b)(iii)

$$E = \frac{dN\phi}{dt} = \frac{dNBA}{dt} = NA \left(\frac{dB}{dt} \right)$$

$$E = NA \left(\frac{dB}{dt} \right) = NA \left(\frac{dB}{dx/v} \right) = NA \left(\frac{dB}{dx} \right) v$$

Consider the coil moving at constant speed towards the end of the bar magnet. The induced emf in the coil will increase as the B-x gradient increases.

From the above, it can be seen that to maintain the same induced e.m.f., the rate of change (or rise) of magnetic flux linkage in the coil must be steady. The instantaneous velocity of the coil should therefore be inversely proportional to the B-x gradient. So, when approaching the magnet and the B-x gradient rises sharply, to mitigate the rise, the speed of the coil must decrease gradually. The combined effect is to make the magnetic flux linkage to rise at a steady rate resulting in a steady emf. See graphs on right.

At the 30.0 mm position, since the B-x slope is gradual, the speed of the coil must be large initially [B1]

When the coil approaches the magnet, the B-x gradient increases at an increasing rate wrt x. Thus, to maintain the same emf, the **speed of the coil must decrease at a decreasing rate**

when moving from 30.0 mm position to the 10.0 mm position.
[B1]

9 (c)(i)

For a rotating coil, $E = NBA\omega \sin(\omega t)$

For e.m.f. to be maximum, $\sin(\omega t) = 1$.

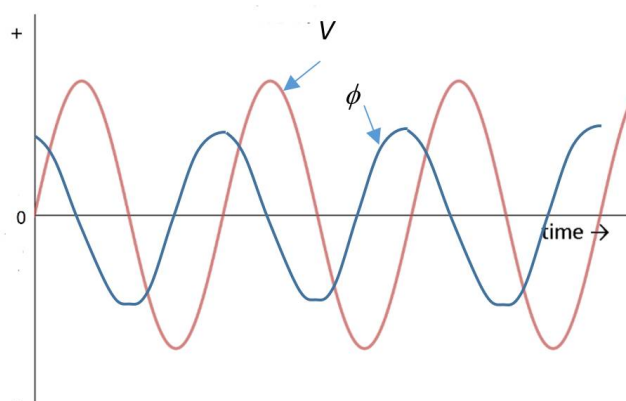
$$\begin{aligned}\text{Hence, max e.m.f. } E_0 &= NBA(2\pi f) \\ &= (800)(0.50)(5.0 \times 10^{-2} \times 8.0 \times 10^{-2})(2\pi)(240/60) \quad [\text{M1}] \\ &= 40.2 \text{ V} \\ &= 40 \text{ V} \quad [\text{A1}]\end{aligned}$$

9 (c)(ii)

$$V = -\frac{d\Phi}{dt} \quad [\text{B1}]$$

Do not award mark if the negative sign is omitted.

9 (c)(iii)



Both graphs have same frequency (period) [B1]

A phase difference of 90° with V lagging ϕ . [B1]

9(c)(iv)

$$I_0 = \frac{V_0}{R} = \frac{40}{12} = 3.3 \text{ A} \quad [\text{A1}]$$

9 (c)(v)

The term r.m.s. value refers to the square root of the mean squared values. [B1]

The r.m.s. current is equivalent to that of a steady direct current which would dissipate energy in a given resistance at the same rate as the alternating current (same heating effect)
[B1]

9 (c)(vi)

$$\begin{aligned}I_{\text{rms}} &= \frac{I_0}{\sqrt{2}} = \frac{3.35}{\sqrt{2}} = 2.37 \text{ A} \\ &= 2.4 \text{ A} \quad [2 \text{ s.f.}]\end{aligned}$$

[A1]

9 (c)(vii)

$$\begin{aligned}\text{Average power} &= I_{rms}^2 R = (2.37)^2 (11.4) = 64.0 \text{ W} \\ &= 64 \text{ W (2 s.f.)} \quad \quad \quad [\text{A1}]\end{aligned}$$